

Netflix - Data Exploration and Visualisation

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1. Problem Statement :

BACKGROUND:

Netflix is one of the world's leading subscription-based video streaming platforms, offering thousands of movies and TV shows across multiple countries. With a vast and diverse content library, understanding the composition of its catalog is essential for making strategic decisions regarding future content production, regional expansion, and content acquisition.

BUSINESS PROBLEM:

Netflix aims to understand the characteristics of its existing content library to identify trends in content type, genre, production country, release timeline, and creator collaborations. By analyzing these patterns, Netflix can make informed decisions about future content investments and identify opportunities for growth in different countries.

Since the available dataset contains only metadata about Netflix titles and does not include performance metrics such as viewership, revenue, watch time, or user ratings, the analysis focuses on identifying historical content patterns rather than measuring content success.

ANALYSIS AIM:

The primary aim of this analysis is to perform Exploratory Data Analysis (EDA) on the Netflix content dataset to uncover meaningful patterns in its catalog. The analysis seeks to:

- Understand the overall composition of Netflix's content library.
- Analyze the distribution of Movies and TV Shows across different countries.
- Examine trends in content releases and additions over time.

- Identify the most common genres, ratings, directors, and actors featured on the platform.
- Explore regional differences in content production.
- Generate data-driven business insights that can support future content planning and international growth strategies.

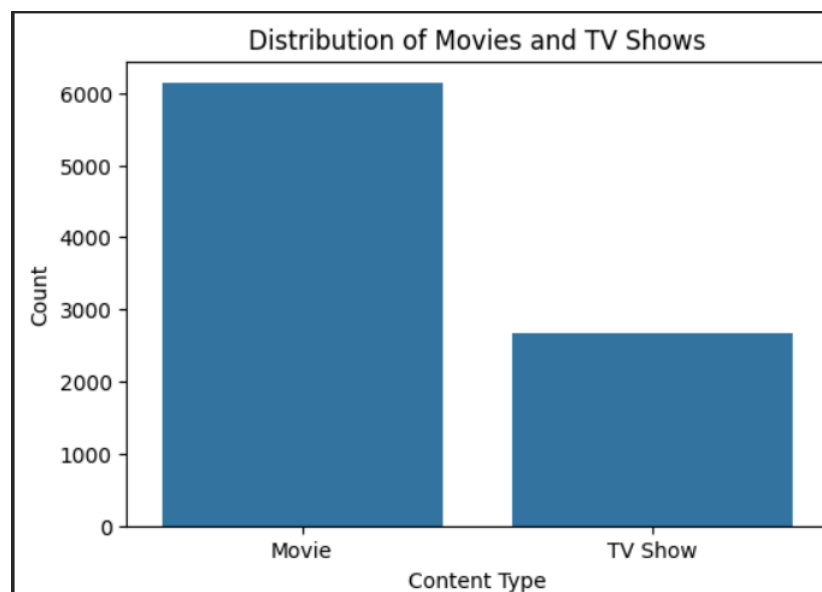
DATASET DESCRIPTION:

- **Show_id:** Unique ID for every Movie / Tv Show
Type: Identifier - A Movie or TV Show
Title: Title of the Movie / Tv Show
Director: Director of the Movie
Cast: Actors involved in the movie/show
Country: Country where the movie/show was produced
Date_added: Date it was added on Netflix
Release_year: Actual Release year of the movie/show
Rating: TV Rating of the movie/show
Duration: Total Duration - in minutes or number of seasons
Listed_in: Genre
Description: The summary description
- Number of movies in the Netflix Dataset: **8,807**

2. Insights on Graphical Analysis:

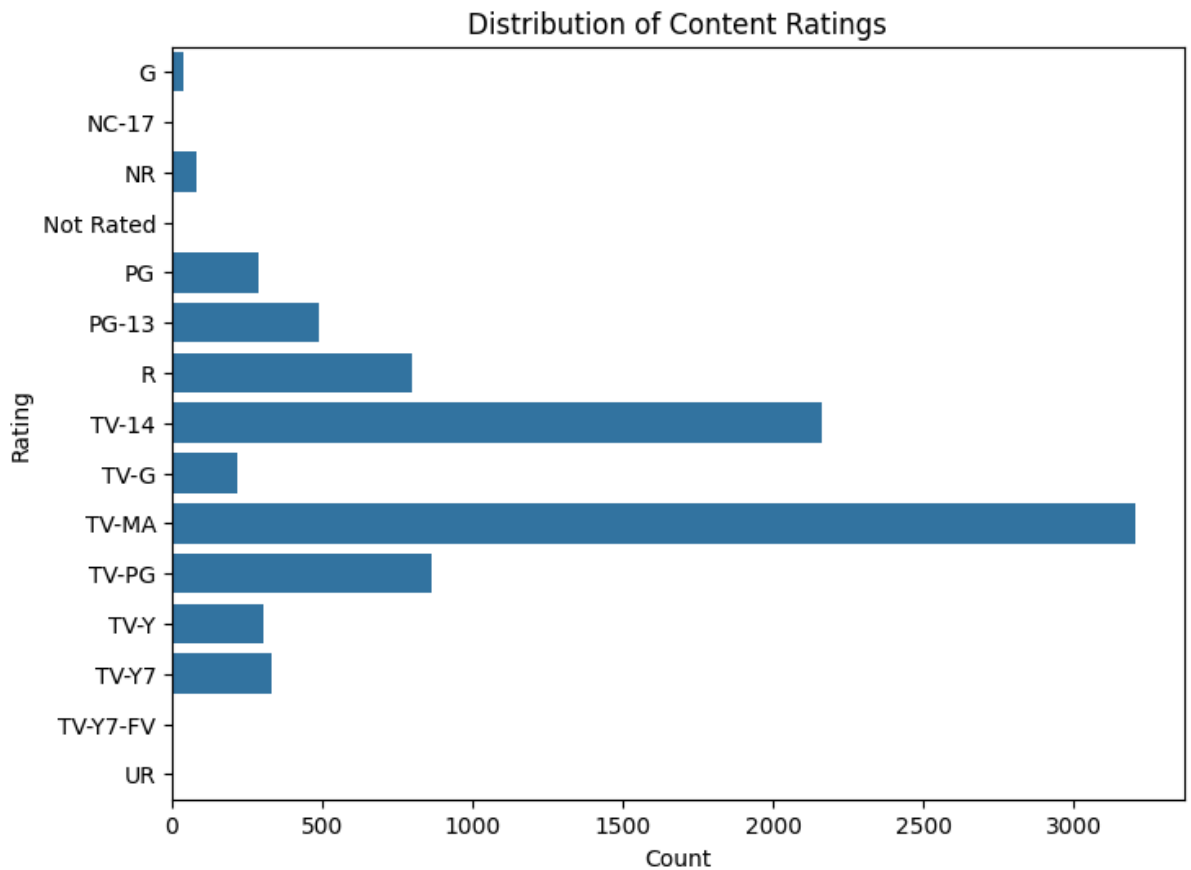
Distribution of Movies and TV Shows:

The dataset contains significantly more movies than TV shows, with movies making up the majority of the content.



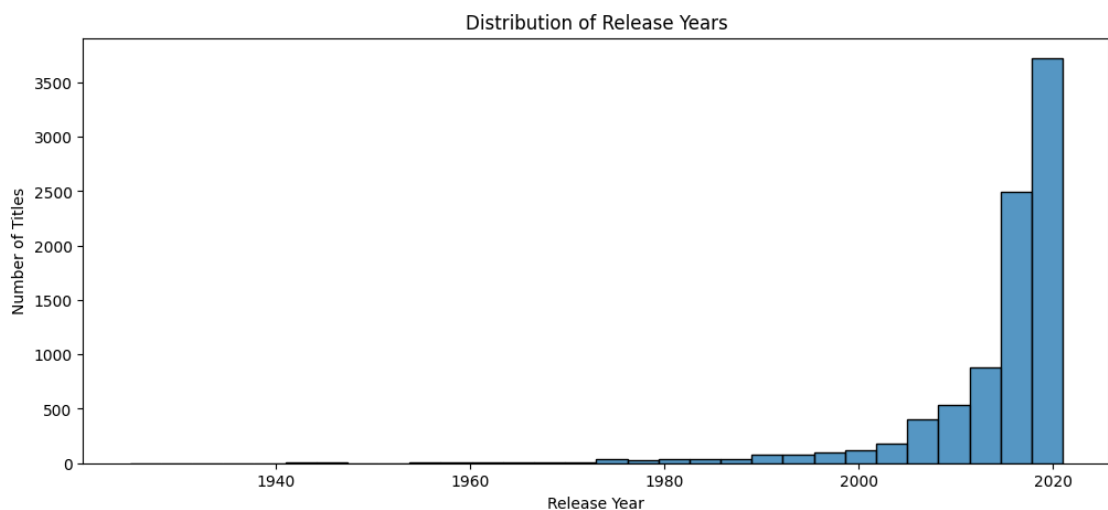
Distribution of Content Ratings:

'TV-MA' and 'TV-14' are the most common content ratings, indicating a prevalence of mature content on the platform. Ratings like 'G', 'NC-17', and 'UR' are less frequent.



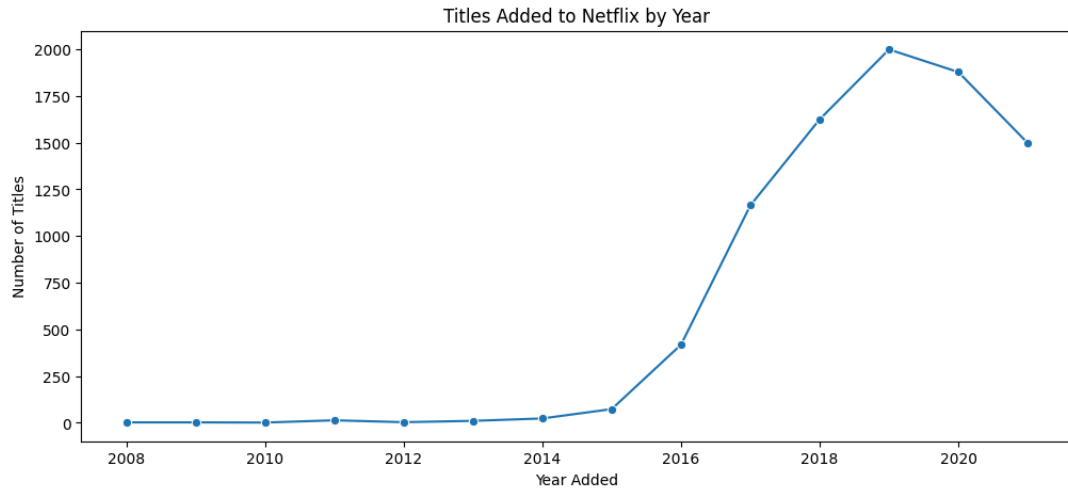
Distribution of Release Years:

The release years for content on Netflix show a clear increase in volume from the early 2000s, with a significant surge in releases from 2010 onwards, peaking around 2017-2019.



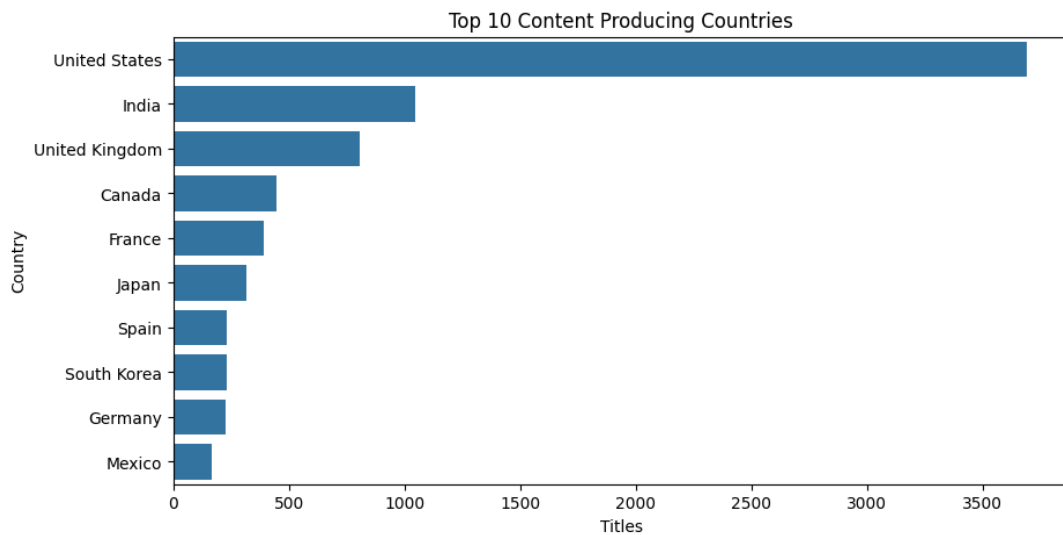
Titles Added to Netflix by Year:

The number of titles added to Netflix steadily increased from 2008, experiencing rapid growth from 2016 and peaking in 2019, before showing a slight decrease in 2020 and 2021.



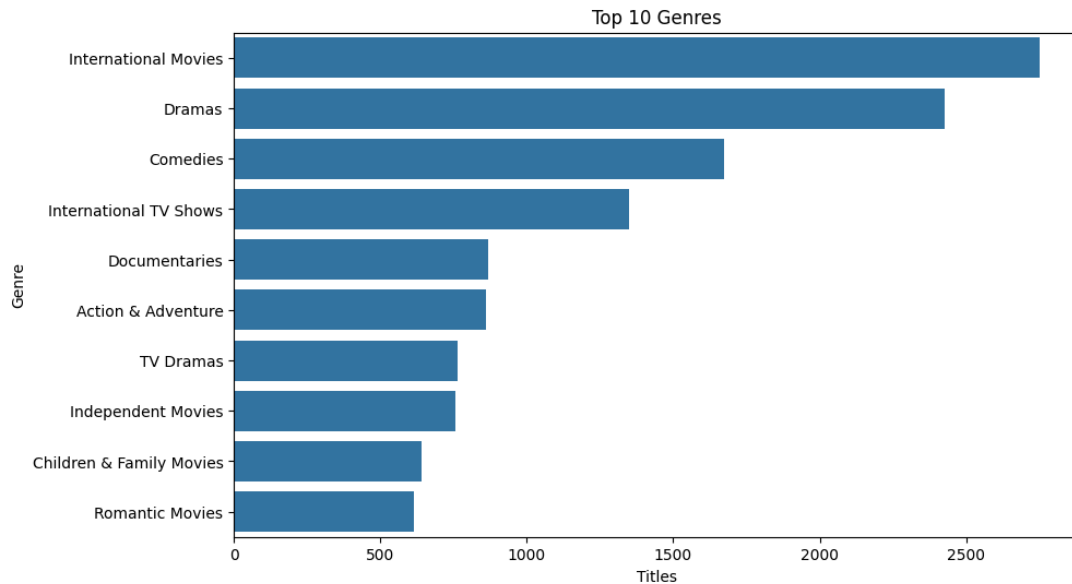
Top 10 Content Producing Countries:

The United States is by far the largest content producer, followed by India and the United Kingdom. Other notable countries include Canada, France, and Japan.



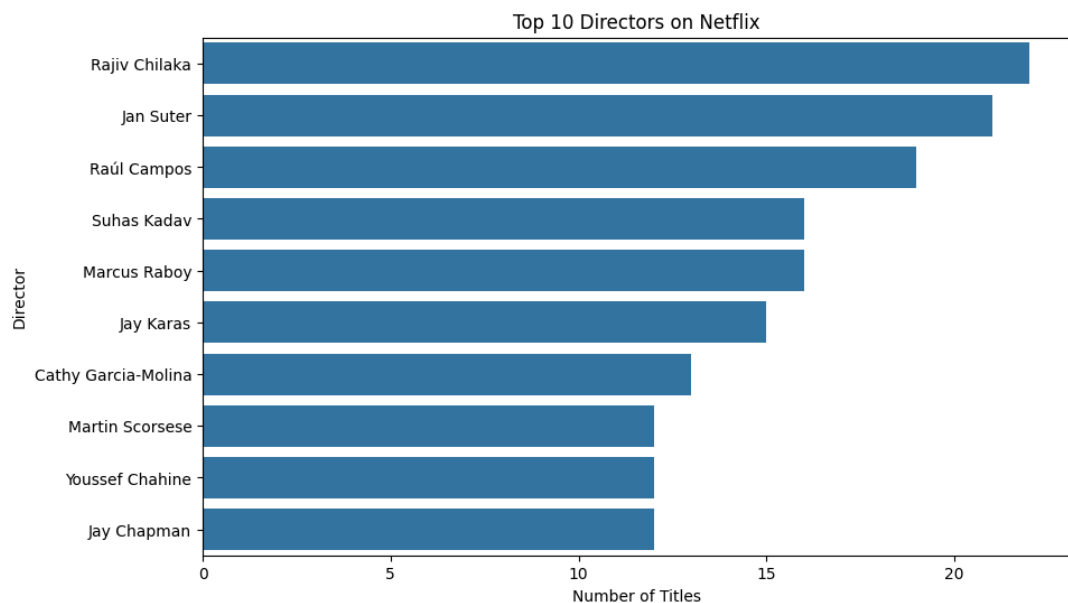
Top 10 Genres:

'International Movies' and 'Dramas' are the most frequently listed genres, followed closely by 'Comedies' and 'International TV Shows'. This highlights a strong focus on both global and dramatic/comedic content.



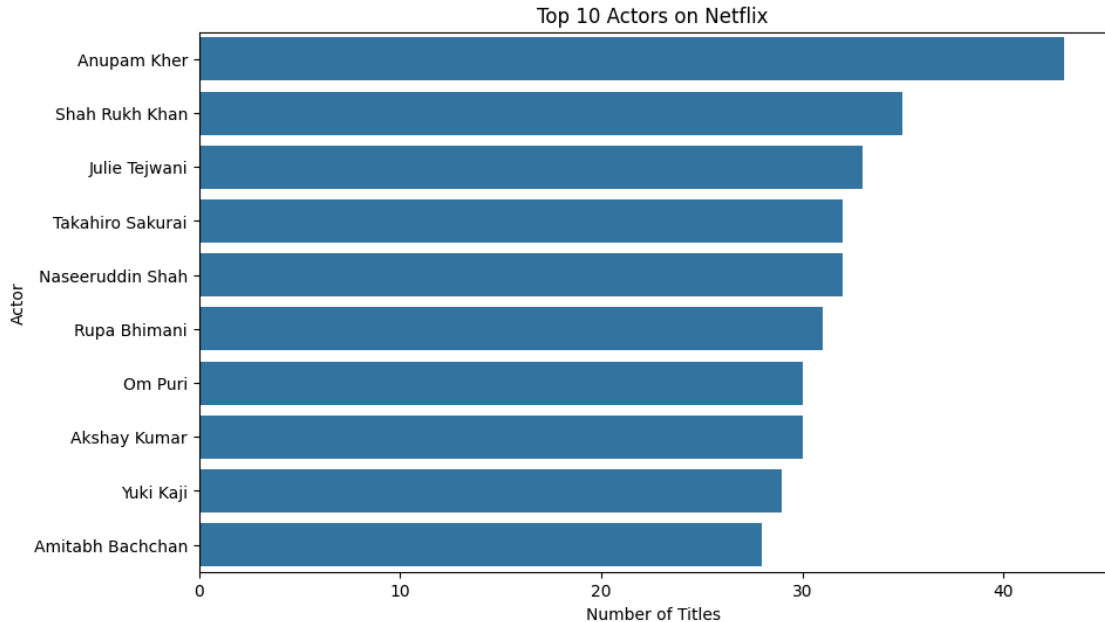
Top 10 Directors on Netflix:

'Rajiv Chilaka', 'Jan Suter', and 'Raúl Campos' are among the most prolific directors on Netflix, with 'Unknown' being the highest category due to missing data.



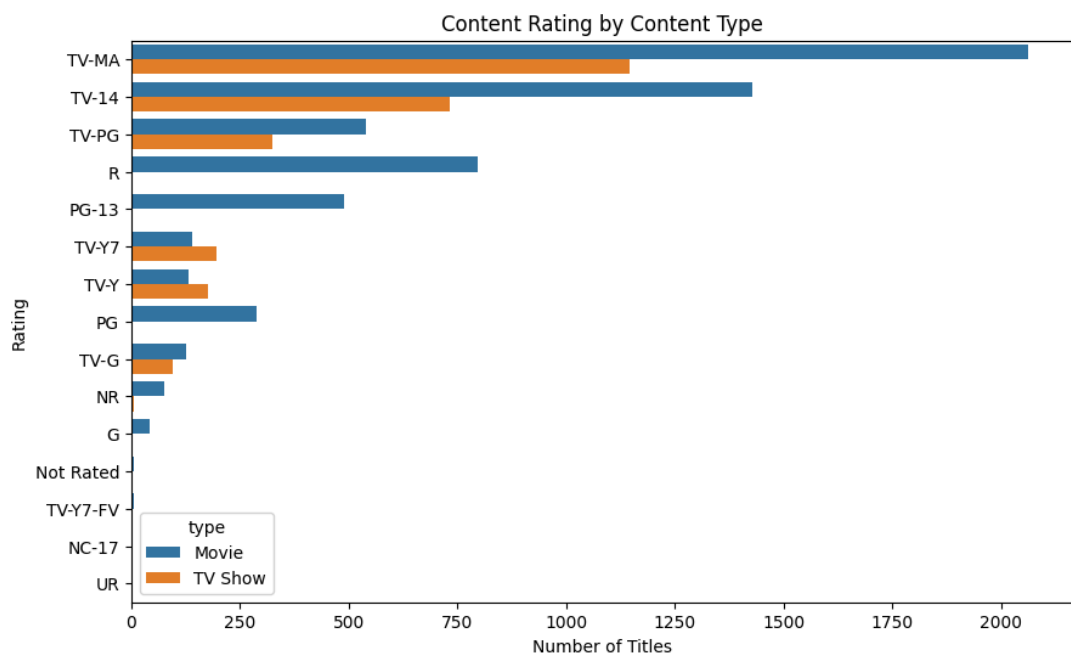
Top 10 Actors on Netflix:

'Anupam Kher' leads as the actor with the most titles, followed by 'Shah Rukh Khan' and 'Julie Tejewani', indicating a strong presence of actors from various regions, particularly India.



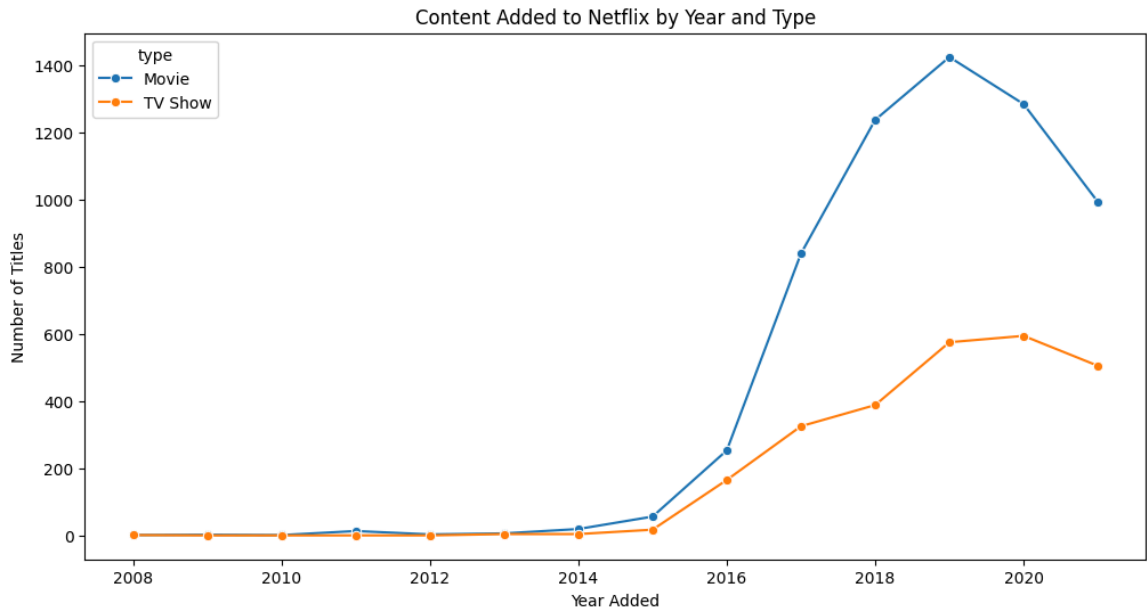
Content Rating by Content Type:

Movies and TV shows have distinct rating distributions. 'TV-MA' and 'TV-14' are dominant for both, but movies show a broader range of ratings including 'R' and 'PG-13', while TV shows are more concentrated in the 'TV' ratings.



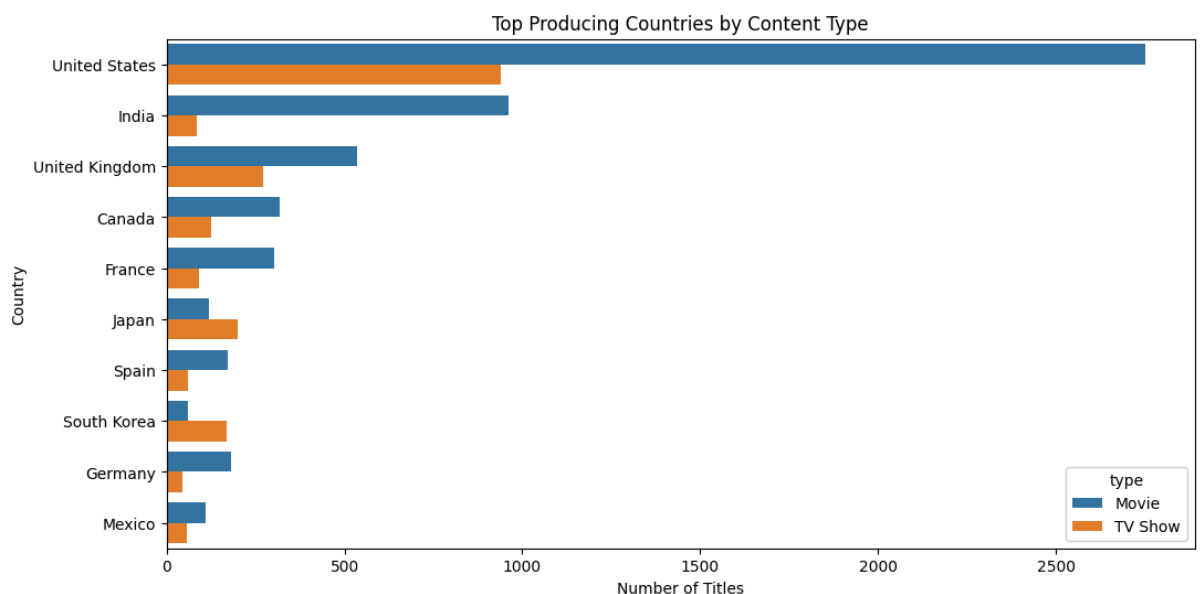
Content Added to Netflix by Year and Type:

The growth in content added to Netflix from 2015 to 2019 was driven by both movies and TV shows, with movies consistently having a higher volume of additions each year. Both content types show a slight decline in additions in 2020 and 2021 compared to 2019.



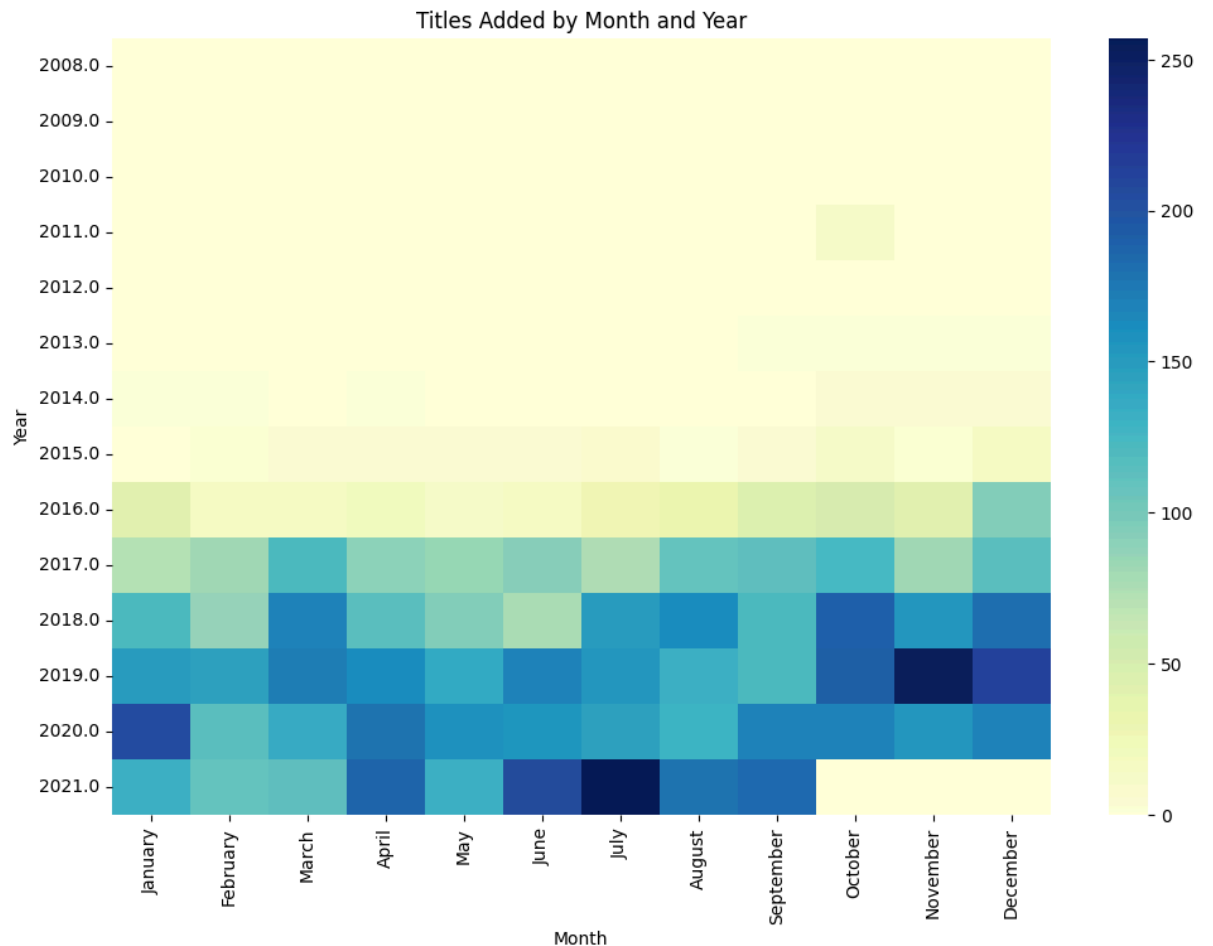
Top Producing Countries by Content Type:

The United States contributes heavily to both movies and TV shows. India is a significant producer of movies, while the UK also shows a strong presence in both categories. Other countries tend to specialize more in one content type, or have a smaller overall contribution.



Titles Added by Month and Year:

The heatmap indicates a general increase in content additions over the years, with specific months like November and December showing higher activity in certain years (e.g., 2019, 2020). There isn't a consistent strong monthly pattern across all years, but late year additions seem common as the number of titles added increases.



3. Insights based on Non-Graphical Analysis:

Comments on the range of attributes:

- **show_id and title:** Both have 8807 unique values, indicating each entry is distinct for show ID and title. This suggests no duplicate show entries based on these identifiers.
- **type:** Contains 2 unique values (Movie, TV Show), confirming the dataset categorizes content into these two main types.
- **director:** Has 4529 unique directors. Given there are 8807 total entries, this indicates many directors have multiple titles, and also highlights the significant number of 'Unknown' directors (2634 in `director_df.value_counts().head(10)`).
- **cast:** There are 7693 unique cast members. Similar to directors, many cast members appear in multiple titles, and 'Unknown' is a common value.
- **country:** Features 749 unique countries, showing the global diversity of content production. 'Unknown' is also a common value here.
- **date_added:** 1699 unique dates when content was added, indicating a varying number of additions over time.
- **release_year:** Content spans 74 unique release years, ranging from 1925 to 2021, showcasing a wide historical coverage.
- **rating:** 15 unique content ratings are present, reflecting the diverse audience segmentation on the platform.
- **duration:** 220 unique duration values, including both 'min' for movies and 'Seasons' for TV shows.
- **listed_in (genre):** 514 unique genre combinations, showing the highly granular categorization of content genres.
- **year_added, month_added, day_added:** 14 unique years, 12 unique months, and 7 unique days of the week for when content was added, which will be useful for temporal analysis.

Comments on the distribution of the variables and relationship between them:

- **Content Type Distribution:** The dataset is heavily skewed towards movies, with 6131 movies compared to 2676 TV shows. This implies Netflix's content library primarily consists of movies.
- **Content Rating Distribution:** 'TV-MA' is the most frequent rating (3207), followed by 'TV-14' (2160), suggesting a large portion of Netflix content is geared towards mature audiences. 'G', 'NC-17', 'UR', and 'Not Rated' are the least common ratings.
- **Content Addition Over Time:** The number of titles added to Netflix significantly increased from 2016, peaking in 2019 (1999 titles), and then slightly decreasing in 2020 and 2021. This indicates a period of rapid expansion in content acquisition.
- **Content Release Year Distribution:** The majority of content released is more recent, with a high concentration of titles released after 2010, and a sharp peak around 2017-2020.
- **Top Countries (Global Distribution):** The 'United States' is by far the top content-producing country (3690 titles), followed by 'India' (1046) and the 'United Kingdom' (806). This highlights the dominance of these countries in Netflix's content library.
- **Top Genres:** 'International Movies' (2752) and 'Dramas' (2427) are the most prevalent genres, indicating a strong focus on global cinema and dramatic storytelling.
- **Top Directors:** Many directors have a relatively small number of titles. Directors like 'Rajiv Chilaka' and 'Jan Suter' appear frequently, but the 'Unknown' category is the largest, indicating substantial missing director data.

Relationship between Type and Rating (Crosstab):

- Movies are more distributed across a wider range of ratings (e.g., G, PG, PG-13, R, NC-17, UR) in addition to TV ratings.

- TV Shows are almost exclusively rated with 'TV' categories (TV-MA, TV-14, TV-PG, TV-Y7, TV-Y, TV-G), with very few 'R' rated TV shows.
- Both 'TV-MA' and 'TV-14' are dominant for both movies and TV shows, with TV-MA being particularly high for movies and TV shows.

Relationship between Country and Type (Crosstab):

- The 'United States' produces a large number of both movies and TV shows.
- 'India' shows a strong preference for producing movies over TV shows for Netflix.
- The 'United Kingdom' has a significant contribution to both movies and TV shows, though with more movies.
- Countries like 'Japan' and 'South Korea' also contribute to both types, but their TV show contribution is notably higher relative to other countries like India.

4. Business Insights:

Content Strategy & Focus:

- **Dominance of Movies:** Netflix's content library is overwhelmingly dominated by movies (approx. 69%), suggesting that the platform's core offering and audience preference lean heavily towards film content. Any strategic shifts should consider this established foundation.
- **Focus on Mature Audiences:** The prevalence of 'TV-MA' and 'TV-14' ratings (over 60% combined) indicates a strong focus on content for mature and young adult audiences. This suggests that Netflix targets these demographics more aggressively than younger audiences (e.g., TV-Y, TV-G rated content is significantly less).

- **Genre Trends:** 'International Movies' and 'Dramas' are the most frequent genres. This highlights a successful strategy of appealing to a global audience with diverse dramatic storytelling. Continued investment in these areas is likely to resonate well with the existing subscriber base.

Production & Acquisition Trends:

- **Recent Content Boom:** Netflix significantly increased its content acquisition and production from 2016 onwards, peaking in 2019. This period marks aggressive expansion and likely reflects a strategy to rapidly build out its library to attract and retain subscribers during heightened competition. The slight decline in 2020-2021 might be due to production challenges (e.g., pandemic) or a shift towards quality over quantity.
- **Global Sourcing:** The United States is the primary content producer, but India and the UK are also major contributors. This multi-national sourcing strategy allows Netflix to cater to diverse linguistic and cultural preferences, crucial for its international subscriber base. India, in particular, shows a strong focus on movie production for Netflix.

Content Gaps & Opportunities:

- **Director/Actor Data Gaps:** A significant portion of 'director' and 'cast' entries are marked as 'Unknown', indicating potential data quality issues or a lack of metadata for older/smaller productions. Improving this data could enhance recommendation algorithms and user discovery.
- **Outliers in Release Year:** While the bulk of content is recent, the presence of older films (e.g., from 1925, 1940s) suggests a strategy to offer a classic film library. This could be a differentiator, but their impact on overall viewership should be assessed.

Audience Engagement:

- **TV Show Growth:** While movies dominate, TV shows have also seen consistent growth in additions, especially from 2015-2019.

This suggests a growing interest in episodic content, and strategic investment in TV series remains important for subscriber engagement.

- **Monthly Content Release Patterns:** The heatmap of titles added by month and year doesn't show a consistently strong seasonal pattern across all years. However, certain months like November and December show higher additions in later years (2019, 2020), which could indicate a strategy to release more content towards the end of the year to capture holiday viewership or new subscriber sign-ups.

Competitive Landscape:

- **Diverse Offerings:** The wide array of genres, countries, and ratings indicates Netflix's strategy to appeal to a broad global audience, which is essential in a competitive streaming market. Understanding which specific niches are underserved or overserved can inform future content decisions.

5. Recommendations:

1. Continue Investing in Movie Content: Since movies are the dominant content type and have a high acquisition rate, maintain a strong focus on acquiring and producing diverse film content. This aligns with the current audience preference and library strength.

2. Prioritize Mature Audience Content: Given the high volume of 'TV-MA' and 'TV-14' rated content, continue to target and acquire content for adult and young adult demographics. Explore if there are underserved niches within these mature ratings.

3. Expand Global & Drama/Comedy Offerings: Further invest in 'International Movies' and 'Dramas', as these are highly popular genres. Look for opportunities to acquire more content from top-producing countries like India and the UK, and explore emerging markets.

4. Strategic Content Release Schedule: Analyze if there are specific times of the year (e.g., late year, holiday seasons) that consistently show higher content additions. Plan major content releases during these periods to maximize viewership and subscriber engagement.

5. Improve Metadata Quality for Directors and Cast: Implement processes to reduce 'Unknown' entries for directors and cast members. Accurate metadata can enhance content discoverability and improve recommendation algorithms, leading to better user experience.

6. Assess ROI of Older Content: While some older titles are present, evaluate their performance and cost-effectiveness. Understand if a classic film library is a significant draw for subscribers or if resources should be reallocated to newer content.

7. Monitor TV Show Growth: Although movies lead, TV shows are a growing segment. Continue to invest strategically in TV series, especially in popular international TV show categories, to cater to evolving viewing habits and maintain a balanced library.

8. Explore Younger Audience Niche: If business goals include expanding into younger demographics, identify content gaps in 'TV-Y' or 'TV-G' rated categories and strategically acquire or produce content to fill these gaps without alienating the core mature audience.